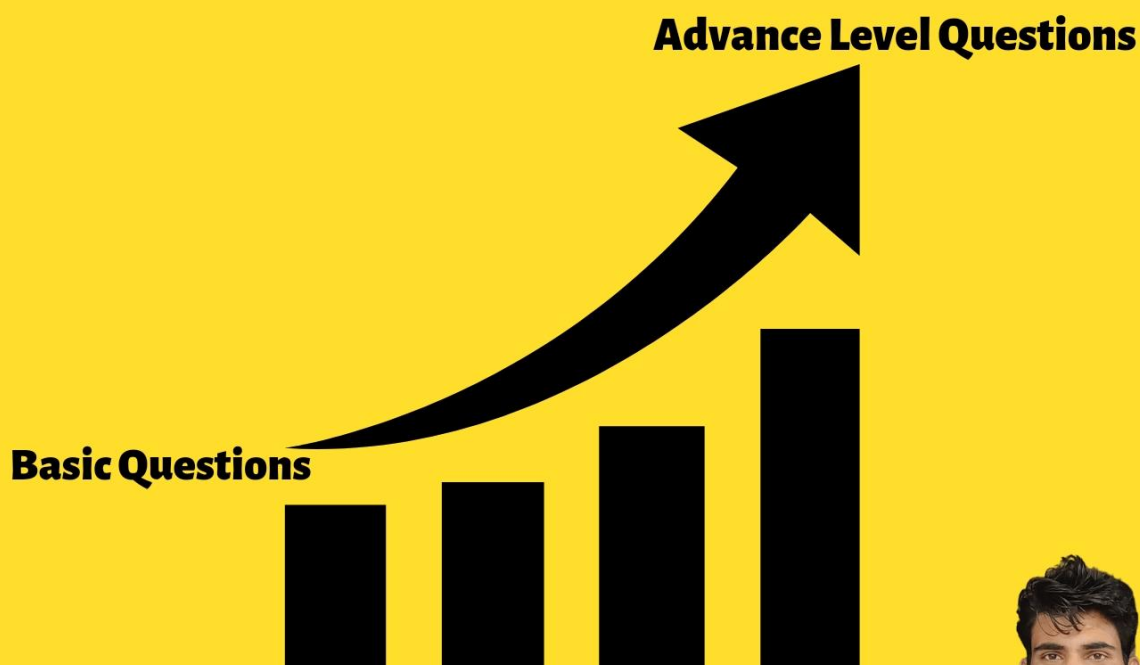


MATHS

PARTNERSHIP

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Partnership

$$\text{Profit} \propto \text{Investment} \times \text{Time}$$

$$\text{Profit} = I_1 T_1 : I_2 T_2 : I_3 T_3 : I_4 T_4 \dots$$

$I_1, I_2, I_3 \dots$ = Investment

$T_1, T_2, T_3 \dots$ = Time

	A	B	C
Capital/Invest	2000	4000	10,000
Time	12 month	9 month	2 month
Profit Ratio:-	2000×12	4000×9	$10,000 \times 2$
	= 6 : 9 : 5		

MathswithGajanand

m-II

	A	B	C
Invest	2000	4000	10,000
Time	12m	9m	2m
Profit Ratio	12	18	10
	6 : 9 : 5		

1 Three partners shared the profit in a business in the ratio $8:7:5$, They invested their capitals for 7 months, 8 months and 14 months, respectively. What was the ratio of their capitals?

Soln:- Basic
m-I

Capital :- $x : y : z$

Time :- $7 : 8 : 14$

Profit :- $\frac{7x}{8} : \frac{8y}{7} : \frac{14z}{5}$

$$\Rightarrow \frac{x}{64} : \frac{y}{49} : \frac{z}{20} \quad \underline{\underline{\text{Ans}}}$$

m-II

$$\begin{aligned} \text{Capital} &= \frac{\text{Profit}}{\text{Time}} = \frac{8}{7} : \frac{7}{8} : \frac{5}{14} \\ &= \boxed{64 : 49 : 20} \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

2 A, B and C enter into a partnership with capitals in the ratio $5:6:8$. At the end of the business term, they received the profit in the ratio $5:3:12$. Find the ratio of time for which they contributed their capitals?

Soln:-

$$\begin{aligned} \text{Time} &= \frac{\text{Profit}}{\text{Capital}} = \frac{5}{5} : \frac{3}{6} : \frac{12}{8} \\ &= 1 : \frac{1}{2} : \frac{3}{2} \\ &= \boxed{2 : 1 : 3} \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

[3] A, B and C invested capital in the ratio 5 : 7 : 4, the timing of their investments being in the ratio $x : y : z$. If their profits are distributed in the ratio 45 : 42 : 28, then $x : y : z = ?$

[SSC CGL Mains - 2020]

Soln:-

$$x : y : z$$

$$\frac{45}{5} : \frac{42}{7} : \frac{28}{4}$$

$$\boxed{9 : 6 : 7} \text{ Ans}$$

[4] X, Y and Z are partner in a business. If X's capital is twice of Y's capital and Y's capital is three times to that of Z's capital then find the ratio of their investments:

Soln:- $X = 2Y$, $Y = 3Z$

$$X : Y : Z$$

$$\begin{array}{r} 2 : 1 : 1 \\ \swarrow \quad \searrow \\ 3 : 1 : 1 \\ \hline 6 : 3 : 1 \end{array} \text{ Ans}$$

[5] A, B and C started a business. Twice the investment of A is equal to thrice the investment of B and also five times the investment of C. If the total after a year is Rs. 15.5 lakhs, then the share of B in the profit is (in lakhs)

Soln:-

$$2A = 3B = 5C$$

$$A : B : C$$

$$15 : 10 : 6 \rightarrow 31$$

$$B = \frac{15.5}{31} \times 10$$

$$= \boxed{5 \text{ lakhs}} \text{ Ans}$$

- 6] The ratio of investment by A to that by B in a business is 14:15 and the ratio of their respective profits at the end of a year is 2:5. If A invested the money for 3 months, then for how much time (in months) B invested his money?

Soln:-

$$\text{Time} = \frac{P}{C} = \frac{2}{14} : \frac{5}{15}$$

$$\xrightarrow{3} 3 : 7 \xrightarrow{\quad} \boxed{7 \text{ month}} \quad \underline{\underline{\text{Ans}}}$$

- 7] The ratio of the investments of A and B in a business is 7:5, and the ratio of their profits at the end of a year is 2:5. If A invested the money for 6 months, then for how much time (in months) has B invested his money?

Soln:-

$$\text{Time} \rightarrow \frac{2}{7} : \frac{5}{5}$$

$$\xrightarrow{\times 3} 2 : 7 \xrightarrow{\times 3} \boxed{21 \text{ month}} \quad \underline{\underline{\text{Ans}}}$$

- 8] X and Z invest capital in the ratio of 2:1 while X and Y invest capital in the ratio of 3:2. If their annual Profit is ₹ 1,57,300 then what is Y share?

Soln:-

$$\begin{array}{ccc} X & Y & Z \\ 3 \times 2 & & 3 \times 1 \\ 2 \times 3 & 2 \times 2 & \\ \hline 6 & 4 & 3 \end{array} \leftarrow \text{Profit}$$

$$Y = \frac{157300}{13} \times 4$$

$$Y = 12100 \times 4$$

$$Y = \boxed{48400} \text{ Ans}$$

[9] A, B and C invested their capitals in the ratio 2:3:5. The ratio of months for which they invested is 4:2:3, respectively. If the difference between the profit shares of A and B is Rs. 1,86,000, then C's share of profit (in Rs) is!

Soln:-

I →	A	:	B	:	C
	2		3		5
T →	4	:	2	:	3
	8	:	6	:	15
Profit →	8	:	6	:	15

(2)

$$2 \rightarrow 1,86,000$$

$$1 \rightarrow 93000$$

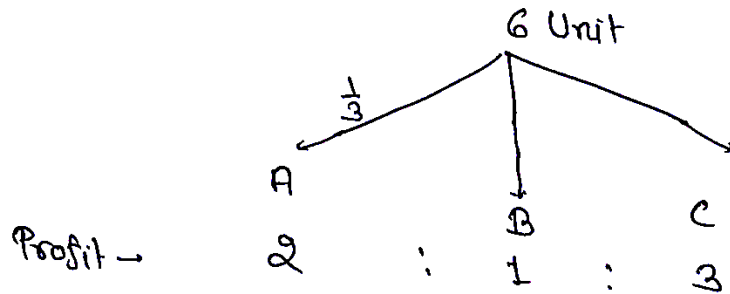
$$C = 15 \rightarrow 15 \times 93000$$

$$= \boxed{1395000} \text{ Ans}$$

[10] A, B and C start a business. A invests $33\frac{1}{3}\%$ of the total capital, B invests 25% of the remaining and C invests the rest. If the total profit at the end of a year is ₹1,62,000 then A's share in profit is!

Soln:-

Let Total Invest



$$A = \frac{147000}{6} \times 2 = \boxed{54,000} \text{ Ans}$$

- [12] Three partner A, B and C started a business by investing Rs. 48,000 each. after 6 months, A left the business after 10 months, B left the business and after 12 month C left the business. If total earned profit is Rs. 5250, then find the share of A, B and C?

Soln:-

	A	B	C
Capital →	48000	48000	48000
	1	1	1
Time →	6	10	12
Profit →	6	10	12
	3	5	6
	+ 14 Unit → 5250		
	1 Unit → 375		

$$A = 375 \times 3 = 1125$$

$$B = 375 \times 5 = 1875$$

$$C = 375 \times 6 = 2250 \text{ Ans}$$

- 13] A, B and C jointly start a business A puts in ₹15,000 for 8 months, B puts in ₹12,000 for 9 months and C puts in ₹8,000, for the whole year. In the end of the year there is a profit of ₹10,800. The difference between A's share and C share in the profit will be?

Soln:-

	A	B	C
Capital	15000	12000	8000
Time	8	9	12
	<u>120</u>	<u>108</u>	<u>96</u>

Profit → 10 : 9 : 8 ⇒ 27 Unit → 10,800

1 Unit → 400

2 Unit → 800 Ans

- 14] X and Y take a grass ground on lease for ₹300 for grazing their animals. If X grazes 10 animals for 5 weeks and Y grazes 15 animals for 7 weeks. The ratio in which they should divide the rent is:

Soln:-

Total Rent = ₹300.

	X	Y
No. of Animals	10	15
Time	<u>5</u>	<u>7</u>
	50	105

10 : 21 Ans

[15] A, B and C started a business investing amounts of Rs. 13,750, Rs. 16,250 and Rs. 18,750 respectively. If B's share in the profit earned by them is Rs. 5,200, what is the difference in the profit (in Rs.) earned by A and C?

[SSC CPO-2020]

Solⁿ:-

	A	B	C
Investment	13750	16250	18750
÷ 25	55	65	75
Ratio	11	13	15

$13 \text{ Unit} \rightarrow 5200$
 $1 \text{ Unit} \rightarrow 400$
 $4 \text{ Unit} \rightarrow 400 \times 4 = 1600$

$A - C = 11 - 15 = -4$
 $A - C = 4 \text{ Unit} \rightarrow 400 \times 4 = 1600$

Ans

[16] A, B and C together invest Rs. 53,000 in a business. A invests Rs. 5,000 more than B and B invests Rs. 6,000 more than C. Out of a total profit of Rs. 31,800, find the share of A.

Solⁿ:-

	A	B	C
Investment	$x + 11000$	$x + 6000$	x

$$3x + 17000 = 53000$$

$$3x = 36000$$

$$x = 12000$$

	A	B	C
Investment	23000	18000	12000
Ratio	23	18	12

$\Rightarrow \text{Total} = 53$

$$A = \frac{31800}{53} \times 23 = 13800$$

Ans

17 A sum of Rs. 8,200 was divided among A, B and C in such a way that A had Rs. 500 more than B and C had Rs. 300 more than A. How much was C's share (in Rs)?

Soln:-

A	B	C
$x+500$	x	$x+800$

$$3x + 1300 = 8200$$

$$3x = 6900$$

$$x = 2300$$

$$C = x + 800 = 2300 + 800 = \boxed{3100} \text{ Ans}$$

18 X, Y and Z enter into partnership. X invests $\frac{1}{4}$ part of total capital for one-fourth of the time. Y contributes one-fifth of the capital for half of the time. Z contributes the remaining capital for the whole time. How should they divide a profit of ₹ 1140?

Soln:-

Let the total time = 8 years ($\frac{1}{4} \times \frac{1}{2} \Rightarrow 8$)

Let the total capital = 20 Units ($\frac{1}{4} \times \frac{1}{5} \Rightarrow 20$)

	X	:	Y	:	Z
Capital $\rightarrow$	5	:	4	:	11
Time $\rightarrow$	<u>2</u>	:	<u>4</u>	:	<u>8</u>
Profit $\rightarrow$	10	:	16	:	88

$$\Rightarrow 5 : 8 : 44 \Rightarrow 57 \text{ (Total)}$$

$$57 \text{ Unit} \rightarrow 1140$$

$$1 \text{ Unit} \rightarrow 20$$

$$X = 20 \times 5 = \boxed{100}$$

$$Y = 20 \times 8 = \boxed{160}$$

$$Z = 20 \times 44 = \boxed{880} \text{ Ans}$$

19] A, B and C become partners in a business. A contributes $\frac{1}{3}$ rd of the capital for $\frac{1}{4}$ th of the time. B contributes $\frac{1}{5}$ th of the capital for $\frac{1}{6}$ th of the time and C the rest of the capital for the whole time. If the Profit is ₹1,820, then the A's share in profit is:

Soln:- Let total Capital = 15 Unit ($\frac{1}{3}, \frac{1}{5} \Rightarrow \text{LCM of } (3, 5) = 15$)
 Let total time = 12 Yr ($\frac{1}{4}, \frac{1}{6} \Rightarrow \text{LCM of } (4, 6) = 12$)

	A	B	C
Capital $\rightarrow$	5	3	7
time $\rightarrow$	$\frac{3}{15}$	$\frac{2}{6}$	$\frac{12}{84}$
Profit $\Rightarrow$	5	: 2	: 28
	$\Rightarrow$ Total 35 Unit		

$$35 \text{ Unit} \rightarrow 1820$$

$$1 \text{ Unit} \rightarrow \frac{1820}{35} = ₹52$$

$$A = 5 \text{ Unit} \rightarrow 52 \times 5 = \boxed{₹260} \text{ Ans}$$

20] A and B entered into a partnership with certain investments. At the end of 8 months, A withdrew and collected back his money. A and B received profit in the ratio 5:9 at the end of the year. If B had invested Rs. 36,000, then how much (in Rs) had A invested?

Soln:-

$$\begin{aligned}\text{Profit} &\rightarrow A : B \\ &5 : 9 \\ \text{time} &\rightarrow 8\text{month} : 12\text{month} \\ &\Rightarrow 2 : 3\end{aligned}$$

$$\text{Capital} = \frac{\text{Profit}}{\text{Time}} = \frac{A}{2} : \frac{B}{3}$$

$$\begin{aligned}&= 5 : 6 \\ &\swarrow \times 6000 \quad \searrow \times 6000 \\ &\boxed{30,000} \quad 36,000 \text{ Given}\end{aligned}$$

Ans

Q1 In a business A and B gained some amount in a certain ratio. B and C received the profit in the ratio as that of A and B. If A received ₹6,400 and C received ₹10,000. Find the share of B.

Soln:-

$$\begin{array}{ccc}\boxed{A} & \boxed{B} & \boxed{C} \\ a & : & b \\ & \nearrow & \searrow \\ & a & : & b\end{array}$$

$$\text{Profit} \rightarrow a^2 : ab : b^2$$

$$\text{Given } a^2 = 6400$$

$$a = 80$$

$$\text{and } b^2 = 10,000$$

$$b = 100$$

$$\boxed{B} = ab = 80 \times 100$$

$$= \boxed{8000} \quad \underline{\underline{\text{Ans}}}$$

22 A, B and C share profit in the ratio of $\frac{1}{4} : \frac{1}{6} : \frac{7}{12}$.

If C retires, they share the profit of C in the ratio of 4:5 respectively. The new profit sharing ratio of A and B will be:-

Soln:-

A	B	C
$\frac{1}{4} \times 12$	$\frac{1}{6} \times 12$	$\frac{7}{12} \times 12$
3	2	7

$\text{LCM}(4, 6, 12) = 12$

	A	B
New Profit	$3 + \frac{7}{(4+5)} \times 4$	$2 + \frac{7}{(4+5)} \times 5$
	$= 3 + \frac{28}{9}$	$= 2 + \frac{35}{9}$
	$= \frac{55}{9}$	$= \frac{53}{9}$

$\Rightarrow \boxed{55 : 53} \text{ Ans}$

23 A and B started a business investing amounts in the ratio of 2:3. If A has invested an additional amount of ₹10,000, their ratio of investment would have been 3:2, The amount invested by A was:

Soln:- **Ratio method**

A : B
 $2_{x_2} : 3_{x_2}$

$3_{x_3} : 2_{x_3}$

$\Rightarrow 5 \begin{pmatrix} 4 \\ 9 \end{pmatrix} : 6$

5 Unit $\rightarrow$ 10,000

1 Unit $\rightarrow$ 2000

A = 9 Unit $\rightarrow \boxed{18000} \text{ Ans}$

Type

- 24** Ramesh started a business investing a sum of Rs. 40,000. 6 months later, Kevin joined by investing Rs 20,000. If they make a profit of Rs. 10,000 at the end of the year, how much is the share of Kevin? [SSC CGL Mains-2020]

Soln:-

	Ramesh	Kevin
Capital →	40,000	20,000
Time →	$\frac{12 \text{ month}}{2}$	6 month
Profit →	4	1

$$\text{Kevin} = \frac{10,000}{(4+1)} \times 1 = \boxed{\text{Rs. 2000}} \text{ Ans}$$

- 25** A invested an amount of ₹25,000 and started a business. B joined him after one year with an amount of ₹30,000. After two years from starting the business, they earned the profit of ₹46,000. What will be B's share in the profit?

Soln:-

	A	B
C →	$\frac{25,000}{5}$	$\frac{30,000}{6}$
T →	$\frac{2 \text{ Year}}{1}$	$\frac{1 \text{ year}}{1}$
P →	10	6

$$\begin{aligned} \text{B's share} &= \frac{46000}{(10+6)} \times 6 \\ &= \boxed{\text{₹17,250}} \text{ Ans} \end{aligned}$$

Q26 A starts business with a capital of ₹14,000. 5 months later B joins and further two months later C joins them. If the profit sharing ratio in the end of year is 4:3:2, then the money invested by C was:

Soln:-

	A	B	C
Capital →	14000	x	y
Time →	<u>12 month</u>	<u>7 month</u>	<u>5 month</u>
Profit	$\frac{1,68,000}{4}$	$= \frac{7x}{3}$	$= \frac{5y}{2}$
Given →	4	3	2

$$42000 = \frac{5y}{2} \Rightarrow y = \boxed{16,800} \text{ Ans}$$

Q27 A starts a taxi service by investing Rs. 25 lakhs. After 3 months, B joins the business by investing Rs. 40 lakhs then 4 months after B joined, C too joins them by investing Rs. 50 lakhs. One year after A started the business they make Rs. 2,73,000 in profit. What is C's share of the profit?

Soln:-

	A	B	C
Capital →	25	40	50
Time →	<u>12m</u>	<u>9m</u>	<u>5m</u>
Profit →	60	72	50

$$30 : 36 : 25 \Rightarrow 91 \text{ Unit} \rightarrow 2,73,000$$

$$1 \text{ Unit} \rightarrow 3000$$

$$C's \text{ share} = 25 \text{ Unit} = 3000 \times 25$$

$$= \boxed{75,000} \text{ Ans}$$

28 Person A started a business by investing Rs. 65,000.

After a few months, B joined him by investing Rs. 50,000.

Three months after the joining of B, C joined the two with an investment of Rs. 55,000. At the end of the year, A got 50% of profit as his share. For how many months did A alone finance the business?

[CGL Pre-2022]

Soln:-

	A	B	C
Capital →	65000	50000	55000
	13	: 10	: 11
time →	<u>12 month</u>	<u>x+3</u>	<u>x</u>
Profit →	156	: 10(x+3)	: 11x

A got 50% of share and remaining 50% divided in B & C

$$156 = 10x + 30 + 11x$$

$$126 = 21x$$

$$x = 6$$

$$B = 9 \text{ month}$$

$$C = 6 \text{ month}$$

So A alone did business for 3 months Ans

29 A starts a business with ₹45,000. After 6 months B enters in his business with ₹80,000. After one year C invests ₹1,20,000. In what ratio the profit will be divided among A, B and C after two years?

Soln:-

	A	B	C
	45000	: 80,000	: 1,20,000
Capital	9	: 16	: 24

$$\begin{array}{l} \text{Time} \rightarrow \frac{2\text{yr}}{2} : \frac{3\text{yr}}{2} : 1\text{yr} \\ \text{Profit} \rightarrow 18 : 24 : 24 \\ \boxed{3 : 4 : 4} \text{ Ans} \end{array}$$

[30] A started a business by investing Rs. 42,000. After few months B joined by investing Rs. 49,000. If at the end of year A got Rs. 9,000 and B got Rs. 7,000 as a share of their profit. Then after how many months B joined the Business.

Soln:- m-I

	A	B
Capital $\rightarrow$	42000	49000
	6	7
Time $\rightarrow$	12m	(x) $\leftarrow$ let
Profit	72	7x
Given $\rightarrow$	9000	7000
	$\boxed{8 = x}$	
	$\boxed{8 \text{ month}} \text{ Ans}$	

m-II

	A	B
Capital $\rightarrow$	6	7
Profit $\rightarrow$	9	7
Time $\rightarrow$	$\frac{\text{Profit}}{\text{Capital}} = \frac{9}{6} : \frac{7}{7}$	
	$= 3 : 2$	
	$\times 4 \checkmark$	$\times 4 \checkmark$
	12month	$\boxed{8 \text{ month}} \text{ Ans}$

[31] A started a business with Rs. 20,000 and B joined after some time with Rs. 25,000. If A and B share the profit in the ratio of 1:2 respectively, then what is the ratio of the time period of investment for A and B respectively?

Soln:-

	A	B
Capital $\Rightarrow$	20,000	25,000
$\Rightarrow$	4	5

$$\text{Time} = \frac{\text{Profit}}{\text{Capital}} = \frac{1}{4} : \frac{2}{5} \Rightarrow \boxed{5:8} \text{ Ans}$$

[32] A started a business with a capital of Rs. 54,000. and admitted B and C after 4 months and 6 months, respectively. At the end of the year, the profit was divided among the three in the ratio 1 : 4 : 5. What is the sum (in Rs) of the capitals invested by B and C ?

Soln:-

	<u>A</u>	<u>B</u>	<u>C</u>
Profit $\rightarrow$	1	4	5
Time $\rightarrow$	<u>12m</u>	<u>8m</u>	<u>6m</u>
Capital $\rightarrow$	$\frac{1}{12}$	$\frac{4}{8}$	$\frac{5}{6}$
	$\Rightarrow 2$	12	20

$\swarrow \times 27000$
 54000

$\underbrace{\quad\quad\quad}_{+}$
 32 Unit $\rightarrow 32 \times 27000$
 $\Rightarrow \boxed{864000}$ Ans

[33] A and B started a partnership business investing in the ratio of 3 : 8. C joined them after 4 months with an amount equal to $\frac{3}{4}$ th of B. What was their profit (in Rs) at the end of the year if C got Rs. 48,000 as his share?

Soln:-

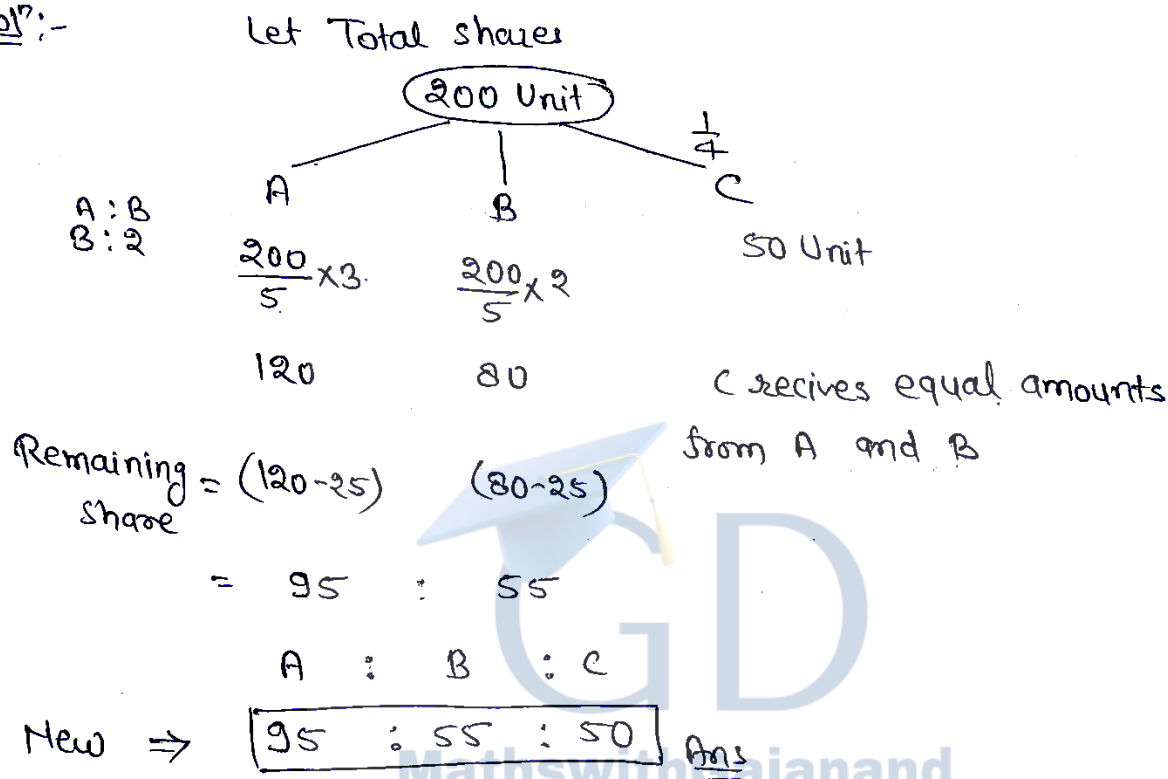
	<u>A</u>	<u>B</u>	<u>C</u>
Capital $\rightarrow$	3	8	6
Time $\rightarrow$	<u>$12m$</u>	<u>$12m$</u>	<u>$8m$</u>
Profit	9	24	12

$\Rightarrow 45$
 $\downarrow \times 4000$
 $\downarrow \times 4000$
 $\downarrow \times 4000$
 48000

$\boxed{1,80,000}$ Ans

34] A and B share profits and losses in a firm in the ratio of 3:2 and C entered in this firm as a new Partner; his profit sharing ratio is $\frac{1}{4}$. If C has taken his share of profit from A and B in equal ratio, then the new profit sharing ratio will be?

Soln:-



35] A and B enter into a partnership with capital in the ratio 5:6. After 4 months, A withdraws $\frac{1}{5}$ of his capital; while B increases his capital by $33\frac{1}{3}\%$. What is the share (in Rs. lakhs) of B in the annual profit of Rs. 6.3 lakhs?

[SSC CGL-Mains-2020]

Soln:- m-I

A	B	
$5 \times 4 \text{ month}$	$6 \times 4 \text{ month}$	
$\downarrow -\frac{1}{5}$	$\downarrow +33\frac{1}{3}\% = +2$	
$4 \times 8 \text{ month}$	$8 \times 8 \text{ month}$	
Profit $\Rightarrow (5 \times 4 + 4 \times 8)$	$(6 \times 4 + 8 \times 8)$	$B = \frac{6.3}{(13+22)} \times 22$ $= \boxed{3.96 \text{ lakh}}$ <u>Ans</u>
$\Rightarrow$	52 : 88	
$\Rightarrow$	13 : 22	

m-II

A

B

$$(5 \times 4 + 4 \times 8) : (6 \times 4 + 8 \times 8)$$

$$13 : 22$$

$$35 \longrightarrow 6.3 \text{ lakh}$$

$$22 \longrightarrow \frac{6.3}{35} \times 22$$

$$\boxed{3.96 \text{ lakh}} \text{ Ans}$$

36 X and Y enter into a partnership with capital in the ratio 3:5. After 5 months X adds 50% of his capital, while Y withdraws 60% of his capital. What is the share of Soln:- X in the annual profit of Rs. 6.84 lakhs?

$$(3 \times 5 + 4.5 \times 7) : (5 \times 5 + 2 \times 7)$$

$$(15 + 31.5) : (25 + 14)$$

$$46.5 : 39$$

Profit $\longrightarrow$

$$\boxed{93 : 78}$$

$$171 \text{ Unit} \longrightarrow 6.84$$

$$\text{MathswithGajanan} \quad X = \frac{6.84}{171} \times 93$$

$$= \boxed{3.72} \text{ Ans}$$

37 A and B started a business in partnership by investing in the ratio of 7:9. After 3 months A withdraw $\frac{2}{3}$ of its investment and after 4 months from beginning B withdraw $33\frac{1}{3}\%$ of its investment. If a total earned profit is Rs. 10,201 at the end of 9 months, find the share of each in profit.

Soln:-

A

B

$$(7 \times 3 + (7 - \frac{2}{3} \times 7) \times 6) : (9 \times 4 + 6 \times 5)$$

$$(21 + 14) : (36 + 30)$$

$$35 : 66$$

$$B = \frac{10201}{101} \times 66 = \boxed{6666}$$

Ans

$$101 \longrightarrow 10201$$

$$A = \frac{10201}{101} \times 35 = \boxed{3535}$$

38] A, B and C started a business with their capitals in the ratio 2:3:5. A increased his capital by 50% after 4 months, B increased his capital by $33\frac{1}{3}\%$ after 6 months and C withdrew 50% of his capital after 8 months, from the start of the business. If the total profit at the end of a year was ₹86,800, then the difference between the share A and C in the profit was:

Soln:-

m-I

A	B	C
$2 \times 4m$	$3 \times 6m$	$5 \times 8m$
$\downarrow +50\%$	$\downarrow 33\frac{1}{3}\%$	$\downarrow -2.5 (50\%)$
$3 \times 8m$	$4 \times 6m$	$2.5 \times 4m$

$$(2 \times 4 + 3 \times 8) : (3 \times 6 + 4 \times 6) : (5 \times 8 + 2.5 \times 4)$$

$$32 : 42 : 50$$

Profit →

$$\boxed{16 : 41 : 25}$$

$$\text{Maths with Gajana} \quad 62 \text{ Unit} \rightarrow 86800$$

$$A - C = 9 \text{ Unit} \rightarrow \frac{86800}{62} \times 9$$

$$= \boxed{12600} \text{ Ans}$$

m-II Short

$$(2 \times 4 + 3 \times 8) : (3 \times 6 + 4 \times 6) : (5 \times 8 + 2.5 \times 4)$$

$$32 : 42 : 50$$

$$16 : 41 : 25$$

$$A - C = \frac{86800}{62} \times 9$$

$$= \boxed{12600} \text{ Ans}$$

[39] A, B and C invested money in the ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{5}$ in a business. After 4 months A doubled his investment and after 6 months B halves his investment. If the total profit at the end of year be Rs. 34,650 then find the share of each in profit.

Soln:-

$$\frac{1}{2} : \frac{1}{3} : \frac{1}{5}$$

A B C

$$\begin{array}{ccc} 15 \times 4m & 10 \times 6m & 6 \times 12m \\ \downarrow & \downarrow & \downarrow \\ 30 \times 8m & 5 \times 6m & \end{array}$$

$$(15 \times 4 + 30 \times 8) : (10 \times 6 + 5 \times 6) : (6 \times 12)$$

$$300 : 90 : 72$$

Profit:-

$$50 : 15 : 12$$

$$77 \text{ Unit} \longrightarrow 34650$$

$$1 \text{ Unit} \longrightarrow 450$$

$$\text{A} = 50 \text{ Unit} \longrightarrow 450 \times 50 = \boxed{22500}$$

$$\text{B} = 15 \text{ Unit} \longrightarrow 450 \times 15 = \boxed{6750}$$

$$\text{C} = 12 \text{ Unit} \longrightarrow 450 \times 12 = \boxed{5400}$$

[40] Three partners A, B and C invested in the ratio of $\frac{5}{4} : \frac{4}{5} : \frac{6}{5}$ in a business. After 3 months A increased his capital by 50%. If the total profit of Rs. 35,700 earned at the end of year, find what was the A's share of profit?

Soln

$$\left(\frac{5}{4} : \frac{4}{5} : \frac{6}{5} \right) \times 20 \text{ Lcm}(4,5,5)$$

$$25 : 16 : 24$$

$$\begin{array}{ccc} A & B & C \\ (25 \times 3 + 37.5 \times 9) : 16 \times 12 : 24 \times 12 \\ 412.5 : 192 : 288 & \Rightarrow & 892.5 \end{array}$$

$$A = \frac{35700}{892.5} \times 412.5 = \boxed{16500} \text{ Ans}$$

[41] A and B enter into partnership investing ₹48,000 and ₹60,000 respectively. After 3 months, A withdraws ₹8,000 while B invests ₹6000 after 6 months of starting of business. Out of the total amount of profit, if A gets ₹12,000 as his share at the end of the year total profit is:

Soln: **m-I**

[A] $48,000 \times 3m$ $\downarrow -8000$ $40,000 \times 9m$	[B] $60,000 \times 6m$ $\downarrow +6000$ $66,000 \times 6m$
--	--

$$(48000 \times 3 + 40,000 \times 9) : (60,000 \times 6 + 66000 \times 6)$$

$$504000 : 756000$$

Profit $\rightarrow$ $\begin{array}{ccc} 2 & : & 3 \\ & + & \\ & 5 & \end{array}$

$$\text{Total profit} = \frac{12000}{2} \times 5 = \boxed{30,000} \text{ Ans}$$

m-II

$$\begin{array}{ccc} A & & B \\ (\overset{8}{48000} \times 3 + \overset{20}{40000} \times 9) : (\overset{3}{60,000} \times 6 + 66000 \times 6) \\ (24 + 60) & : & (60 + 66) \end{array}$$

$$184 : 126$$

$$\boxed{2 : 3}$$

$$\frac{12000}{2} \times 5 = \boxed{30,000} \text{ Ans}$$

42 A and B started a business by investing Rs. 36,000 and Rs. 45,000 respectively. After 4 months B withdraws $\frac{4}{9}$ of his investment, its 5 months After she again invested $\frac{11}{9}$ of its original investment. If the total earned profit at the end of the year, is Rs. 1,17,240, then who will get more money as a share of profit and how much?

Soln:-

A	B	
	$-\frac{4}{9} \times 45000$ $= (-20,000)$	$+\frac{11}{9} \times 45000$ $= +55000$
$36000 \times 12m$	$: (45000 \times 4m + 25000 \times 5m + 80000 \times 3m)$	
36×12	$: (45 \times 4 + 25 \times 5 + 80 \times 3)$	
432	$: (180 + 125 + 240)$	
Profit $\rightarrow$	$432 : 545$ $\quad \quad \quad +$ $977 \text{ Unit} \rightarrow 1,17,240$ $1 \text{ Unit} \rightarrow 120$ $(545 - 432) = 113 \text{ Unit} \rightarrow 120 \times 113$ $= \boxed{13560}$	

B will get Rs 13560 more than A

43 Three partners x, y and z started their business by investing Rs. 40,000, Rs. 38,000 and Rs. 30,000, respectively. After 6 months x and z made additional investments of Rs. 20,000 and Rs. 15,000 respectively. Whereas y withdrew Rs. 8,000. Find the share of y (in Rs) in the total profit of Rs. 38,880 made at the end of the year.

Soln:-

x	y	z
$40,000 \times 6m$	38000×6	$30,000 \times 6m$
$\downarrow +20,000$ $60,000 \times 6m$	$\downarrow -8000$ $30,000 \times 6m$	$\downarrow +15000$ $45000 \times 6m$

X

Y

Z

$$(40,000 \times 6 + 60,000 \times 6) : (38,000 \times 6 + 30,000 \times 6) : (30,000 \times 6 + 45,000 \times 6)$$

$$(40+60) : (38+30) : (30+45)$$

$$100 : 68 : 75 \Rightarrow 243 \text{ Unit}$$

$$243 \text{ Unit} \longrightarrow 38880$$

$$1 \text{ Unit} \longrightarrow 160$$

$$Y = 68 \text{ Unit} \longrightarrow 160 \times 68 = \boxed{10880} \text{ Ans}$$

[44] X and Y entered into a partnership, investing ₹16,000 and ₹12,000 respectively. After 3 months X withdrew ₹5,000 while Y invested 5,000 more. After 3 months more Z joins the business with a capital of ₹21,000. After a year they obtained a profit of ₹26,400. By what amount does the share of Y exceed the share of Z.

Soln:-

X

Y

Z

$$16000 \times 3m$$

$$12000 \times 3m$$

$$21,000 \times 6$$

$$\downarrow -5000$$

$$\downarrow +5000$$

$$11000 \times 9m$$

$$17000 \times 9m$$

$$\text{Profit} \rightarrow 16000 \times 3 + 12000 \times 3 + 21000 \times 6$$

$$11000 \times 9 + 17000 \times 9$$

$$= 147000 : 189000 : 126000$$

$$147 : 189 : 126$$

$$\text{Profit} \rightarrow \boxed{7 : 9 : 6} \Rightarrow 22 \text{ Unit} \longrightarrow 26400$$

$$1 \text{ Unit} \longrightarrow 1200$$

$$3 \text{ Unit} \longrightarrow 3 \times 1200$$

$$= \boxed{3600} \text{ Ans}$$

45] A and B had a joint business in which A invested Rs. 60,000 in the business for one year. After 3 months B invested Rs. 80,000. At the beginning of the second year, A invested Rs. 30,000 more and B withdrew Rs. 5000. At the end of two years, profit earned by A Rs. 35,880. What is the profit earned by B, if they distributed half of the total profit equally and rest in the capital ratio?

Soln:-

A

B

$$(60,000 \times 12m + 90,000 \times 12m) : (80,000 \times 9m + 75,000 \times 12)$$

$$(60 \times 12 + 90 \times 12) : (80 \times 9 + 75 \times 12)$$

$$720 + 1080 : 720 + 900$$

$$1800 : 1620$$

$$100 : 90$$

$$\text{Total Profit} = 190$$

Maths with Gajanand

$$\begin{array}{cc} \text{A} & \text{B} \\ 95 & 95 \\ 47.5 & 47.5 \end{array}$$

$$\begin{array}{cc} 10 : 9 \\ \text{A} & \text{B} \\ 50 & 45 \end{array}$$

$$\begin{array}{cc} \text{A} & \text{B} \\ 47.5 + 50 & 47.5 + 45 \end{array}$$

$$= 97.5 : 92.5$$

$$39 : 37$$

$$\begin{array}{cc} \downarrow \times 920 & \downarrow \times 920 \end{array}$$

$$35880$$

$$\boxed{34040} \text{ Ans}$$

- [46] Two brother invested ₹50,000 and ₹70,000 respectively in a business and agreed that 70% of the profits should be divided equally between them and the remaining profit in the ratio of investment. If one Brothers gets ₹90 more than the other. Find the total profit made in the business.

Soln:-

	I	II	
Capital →	50,000	: 70,000	
	5	: 7	
Profit →	5	: 7	⇒ Total Profit = 12

2 Unit → 90

1 Unit → 45

12 Unit → 45 × 12

30% → 45 × 12

100% → $\frac{45 \times 12}{30} \times 100$

= 1800 Ans

- [47] The investments made by X and Y are in the ratio 3:2. If 5% of total profit is donated and X gets ₹8,550 as his share of profit then what is the amount of total Profit?

Soln:- m-I Let total Profit

100

Donate ↙ 5 ↘

95

3 : 2

$$x = \frac{95}{5} \times 3$$

$$= 57 \rightarrow 8550$$

$$100 \rightarrow \frac{8550}{57} \times 100$$

$$= \span style="border: 1px solid black; padding: 2px;">15000 Ans$$

m-II Profit → 3 : 2

total Profit 5 Unit

$$3 \text{ Unit} \rightarrow 8550$$

$$5 \text{ Unit} \rightarrow \frac{8550 \times 5}{3}$$

$$= 14250$$

$$95\% \rightarrow 14250$$

$$100\% \rightarrow \frac{14250}{95} \times 100$$

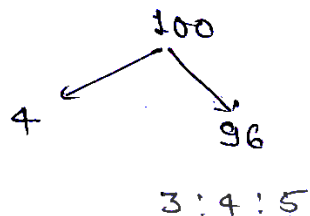
$$= \span style="border: 1px solid black; padding: 2px;">15,000 Ans$$

48 A, B and C are Partners in a firm sharing Profit in the ratio of 3:4:5. If they set aside 4% of the Profits as emergency fund and shared the rest of the profit and B gets his share of Profit as Rs. 1,81,400, the amount of Profit set aside for emergency fund is :-

[SSC CPO-2019]

Soln:-
[m-I]

Total Profit (let)



$$B = \frac{96}{12} \times 4$$

$$= 32 \text{ Unit} \rightarrow 1,81,400$$

$$4 \text{ Unit} \rightarrow \frac{1,81,400}{32} \times 4$$

$$= \boxed{22,675} \text{ Ans}$$

[m-II]

$$\text{Profit} \rightarrow 3:4:5 \Rightarrow 12 \text{ Unit}$$

$$4 \text{ Unit} \rightarrow 1,81,400$$

$$\text{Total Profit} = 12 \text{ Unit} = \frac{1,81,400 \times 12}{4}$$

$$96\% = 5,44,200$$

$$4\% = \frac{5,44,200}{96} \times 4$$

$$= \boxed{22,675} \text{ Ans}$$

49 Two businessmen A and B invest in a business in the ratio 5:8. They decided to reinvest 30% of the profit they earned back into the business. The remaining profit they distributed amongst themselves. If A's share of the Profit was Rs. 87,500 then how much Profit did the business make?

Soln:-

A B
5 : 8

$$5 \text{ Unit} \rightarrow 87,500$$

$$13 \text{ Unit} \rightarrow \frac{87,500 \times 13}{5} = 2,27,500$$

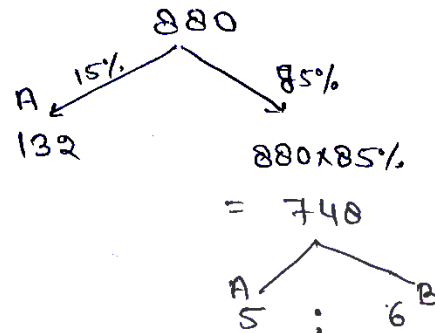
$$70\% \rightarrow 2,27,500$$

$$100\% = \frac{2,27,500}{70} \times 100$$

$$= \boxed{3,25,000} \text{ Ans}$$

[50] A is a Active partner and B is a Inactive partner in business. A puts in ₹5000 and B puts in ₹6,000. A received 15% of the profit for managing the business and the rest is divided in proportion to their capitals. The amount received by A out of the profit of ₹880 in all is:

Soln:-



A : B
5000 : 6000

5 : 6
Profit Ratio

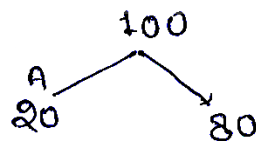
$$A = \frac{748}{11} \times 5 = 340$$

$$\text{Total Profit of A} = 340 + 132 = \boxed{472} \text{ Ans}$$

[51] A, B and C started a business by investing Rs. 27,500 Rs. 32,500 and Rs. 37,500 respectively A is a working Partner and gets 20% of the profit as working allowance and remaining is distributed in the proportion of their investment. If the money received by C is Rs. 13500, what is the total profit?

Soln :-

Let total Profit



$$C = \frac{80}{39} \times 15$$

$$= \frac{400}{13} \rightarrow 13500$$

$$\text{Total Profit} \rightarrow 100 \text{ Unit} = \frac{13500}{4} \times 13$$

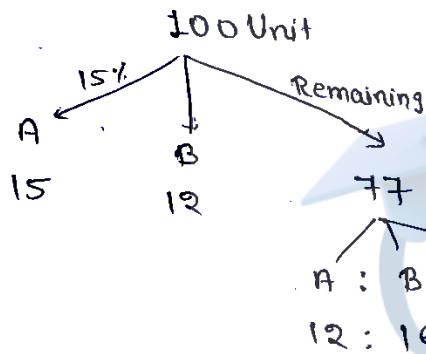
$$= \boxed{43,875} \text{ Ans}$$

A : B : C
Profit → 27500 : 32500 : 37500
11 : 13 : 15
Total = 39

52 A, B and C started a business by investing Rs. 24,000, Rs. 32,000 and Rs. 18,000 respectively. A and B are active Partners and get 15% and 12% of total profit and remaining Profit is to be distributed among them in the ratio of their investment. If C got total Rs. 65,700 as a profit, what was the total amount of profit?

Soln:-

Let total Profit



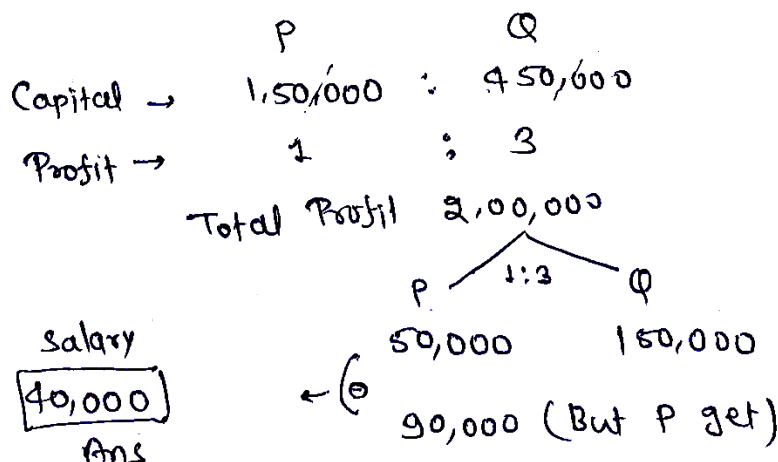
A	B	C
24000	32000	18000
12 : 16 : 9		

$$C = \frac{77}{37} \times 9 \xrightarrow{\text{Given}} 65700$$

$$100 \text{ Unit} \longrightarrow \frac{65700 \times 37}{77 \times 9} = \boxed{370,000} \quad \text{Ans}$$

53 P and Q start a shop with a capital of Rs. 1,50,000 and Rs. 4,50,000, respectively. After a year, out of the profit of 2,00,000, P gets his share of profit plus some money as his salary that is not a part of the profit. If P gets a total of Rs. 90,000, what is the amount of salary that he received?

Soln:-



[54] A, B and C invest in a business in the ratio 4:5:7.

C is a sleeping partner, so his share of profits will be half of what it would have been if he were a working Partner. If they make Rs. 36,000 profit of which 25% is reinvested in the business; how much does B get (in Rs)?

Soln:-

$$\begin{array}{ccc} A & B & C \\ 4 & : & 5 : 7 \end{array}$$

If C is sleeping partner then

$$4 : 5 : \frac{7}{2}$$

$$\Rightarrow 8 : 10 : 7 \Rightarrow 25 \text{ Unit}$$

$$\text{Total Profit} = 36,000$$

$$\begin{array}{l} \swarrow 25\% \quad \searrow 75\% \\ = 36000 \times \frac{75}{100} \\ = 27000 \end{array}$$

$$25 \text{ Unit} \rightarrow 27000$$

$$B = 10 \text{ Unit} \rightarrow \frac{27000}{25} \times 10 = \boxed{10,800} \text{ Ans}$$

[55] A and B start a business by investing equal amounts. Four months later, C joins them investing Rs. 3.5 lakhs. By withdrawing his investment in the business B leaves the business 4 months after C joined. At the end of the year the business makes Rs. 62,400 profit out of which A collects Rs. 24,000 as his share of profit. How much should be paid to C as his share of profit?

Soln:-

$$\begin{array}{ccc} A & B & C \end{array}$$

$$12x : x \times 8m : 3.5 \times 8$$

$$12x : 8x : 28$$

$$12x = 24000$$

$$x = 2000$$

$$B = 8x = 16000$$

$$\begin{array}{ccc} & A & B \\ C = 62400 - 24000 - 16000 \\ = \boxed{22400} \text{ Ans} \end{array}$$

56 Out of total capital required to start a business A invested 30%, B invested $\frac{2}{5}$ th and C invested the remaining capital. At the end of one year sum of Rs. 4,000 is earned as a profit which is 20% of the capital given by B, then find how much C invested in the business?

Soln:- 20% of B's capital = 4000

$$B's \text{ Total Capital} = \frac{4000}{20} \times 100 = 20,000$$

B is $\frac{2}{5}$ of total = 20,000

$$\text{Total Investment} = \frac{20000}{2} \times 5$$

$$= 50,000$$

A	B	C
30%		
= 50,000 × 30%		↓
= 15000	20,000	50,000 - 15000 - 20,000
		15000 <u>Ans</u>

57 A and B started a business in partnership by investing Rs. 10,000 and Rs. 4,000 respectively. Condition of partnership is that B got Rs. 100 per month for management of the business. After paying 5% interest on the capital, annual profit has distributed in the ratio of their investment. Find the share of their profit, if the annual profit is Rs. 9,000.

Soln:- B's profit = 100 × 12 month = 1200

$$\text{Interest of A} = \frac{10,000 \times 5\% \times 1}{100} = 500$$

$$\text{Interest of B} = \frac{4000 \times 5\% \times 1}{100} = 200$$

$$\begin{aligned} \text{Total profit of A \& B} &= 1200 + 500 + 200 \\ &= 1900 \end{aligned}$$

Remaining Profit

$$(4000 - 1900)$$

$$= 2100$$



$$= \frac{2100}{7} \times 5$$

$$= 1500$$

$$\frac{2100}{7} \times 2$$

$$600$$

Total profit of A = 1500
+ 500

2000

Ans

A

B

$$10,000 : 4000$$

5 : 2

Total Profit of B

$$= 600 + 1200 + 200$$

$$= \boxed{2000} \text{ Ans}$$

MathswithGajanand